

Tree Constructions for the ℓ_1 Distortion of Binary Edit Distance

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Abstract

We give two independent constructions of binary words of one length at most d whose ordinary edit-distance metrics require ℓ_1 distortion $\exp(\Omega(\sqrt{\log d \log \log d}))$ for every sufficiently large d . We also prove a constant-distortion binary conversion for one prescribed input length. Together with the companion upper embedding theorem, these lower bounds determine the order of logarithmic distortion uniformly over finite alphabets with at least two symbols.

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1 Introduction

An insertion or deletion changes the positions of every later symbol in a word. A long interval can therefore move at small edit cost even when many positions change. This tension between order and inexpensive shifts is central to the question studied here: how faithfully can one represent the edit distances of an entire set of words by distances in ℓ_1 ?

We use ordinary edit distance $\text{ED}(x, y)$: the minimum number of single-symbol insertions, deletions and substitutions taking x to y , each with cost one. For a finite alphabet Σ with at least two symbols and an integer $d \geq 1$, let $\Sigma^{\leq d}$ be the set of words of length at most d , including the empty word. The cap constrains the words in the domain; an edit script computing their distance may pass through words of any finite length.

For a finite metric space (X, ρ) with at least two points and an injective map $f : X \rightarrow \ell_1$, where ℓ_1 is the space of absolutely summable real sequences, define

$$\text{dist}(f) = \left(\max_{x \neq y} \frac{\|f(x) - f(y)\|_1}{\rho(x, y)} \right) \left(\max_{x \neq y} \frac{\rho(x, y)}{\|f(x) - f(y)\|_1} \right).$$

Let $c_1(X, \rho)$ be the infimum of $\text{dist}(f)$ over such maps, and put $E_\Sigma(d) = c_1(\Sigma^{\leq d}, \text{ED})$. No restriction is imposed on the dimension of the image or on the computation of the map. Throughout the paper, logarithms in asymptotic estimates are natural unless a base is written explicitly.

We prove the following lower bound in two independent ways. Each proof uses a tree to combine inexpensive shifts with a displacement that no alignment can hide.

Theorem 1.1 (Binary lower bound). *There are an absolute constant $c > 0$ and an integer d_0 such that, for every integer $d \geq d_0$, there exist an integer $1 \leq n \leq d$ and a finite set $\mathcal{W} \subseteq \{0, 1\}^n$ with $|\mathcal{W}| \geq 2$ satisfying*

$$c_1(\mathcal{W}, \text{ED}) \geq \exp\left(c\sqrt{\log d \log \log d}\right).$$

The words in a witness all have the same length, although that length may depend on the cap d . At tree depth k , each construction produces logarithmic distortion of order at least $k \log k$ and logarithmic word length of order $k^2 \log k$. Balancing these two scales gives Theorem 1.1 for every sufficiently large cap, not just for a subsequence of caps.

The lower bound is proved here without a companion input. The uniform upper embedding theorem of the finite-circle companion, stated precisely in Section 6, then gives

$$\log E_\Sigma(d) = \Theta\left(\sqrt{\log d \log \log d}\right)$$

with constants independent of the finite alphabet Σ . This is a matching exponential scale, not a constant-factor estimate for $E_\Sigma(d)$ itself.

1.1 History and the embedding question

The single-symbol error model is classical in coding and sequence comparison. Levenshtein studied binary codes correcting insertions and deletions [11], and Wagner and Fischer formulated string correction through increasing partial alignments [17, Sections 3 and 5]. An embedding asks for one set of coordinates that represents every pair in a domain. It is a different task from estimating the distance of one given pair of words. For supplied pairs of binary words, [13, Theorem 1.1] gives a randomized approximation scheme with almost-linear worst-case expected time for each fixed rational accuracy.

Ostrovsky and Rabani proved an ℓ_1 distortion bound $\exp(O(\sqrt{\log n \log \log n}))$ for binary words of a fixed length n [15]. Their recursion compares overlapping substrings and represents their distributions in smaller metric spaces. They also noted extensions to larger alphabets and varying lengths. For the upper bound used at the end of this paper, the companion uniform upper embedding theorem [14, Uniform upper embedding theorem] gives the full quantitative statement on $\Sigma^{\leq d}$ with constants independent of the finite alphabet.

On the lower-bound side, Andoni, Deza, Gupta, Indyk and Raskhodnikova realized complete bipartite graph metrics in binary edit distance and obtained ℓ_1 distortion approaching $3/2$ [1, Section 2 and Theorem 3.1]. Their witness uses three adjacent word lengths. Khot and Naor developed Fourier methods for nonembeddability and obtained an edit-distance lower bound of $(\log n)^{1/2-o(1)}$ for binary words of a fixed length [8]. Their insertion-deletion metric and the ordinary metric used here differ by at most a factor of two. Krauthgamer and Rabani then proved an $\Omega(\log n)$

lower bound for binary words of length n [10, Theorem 1.3]. Khot and Naor used Boolean noise sensitivity, while Krauthgamer and Rabani sharpened the obstruction through an influence inequality for bit flips and cyclic shifts. These cut and shift arguments are methodological predecessors of the proofs here. A cut records which pairs a binary coordinate separates, and Fourier coefficients describe how that separation changes under translations. We prove the particular translation estimates needed for our trees.

Alphabet reduction also has a substantial history. Schulman and Zuckerman use separation between long intervals of distinct codewords in concatenated insertion–deletion codes [16], and synchronization strings protect positions against insertion and deletion errors [5]. More directly, Bhattacharya, Dey, Goldenberg and Koucký state a map on all finite words with an insertion–deletion guarantee normalized by the sum of the word lengths, for pairs of positive total length. The block substitution used to prove their stated theorem has logarithmic width and, for a fixed error parameter, a constant target alphabet [2, Theorem 3; full version, Section 3.2.1 and Lemma 3.3]. For equal-length inputs, their stated normalized guarantee, combined with the elementary delimiter construction below, gives a binary reduction with logarithmic width and absolute distortion. Our lower proofs do not use that result. They use the short delimiter code, whose width-dependent loss is harmless at the tree scale; the appendix gives a separate constant-distortion code for one prescribed word length. For comparison, the isometric Hamming-to-edit embedding of Bhattacharya et al. [3, Theorem 7] starts from a different input metric. The conversions proved here compare edit distance on the source words with edit distance on their binary encodings.

1.2 Two independent tree proofs

Both constructions use an ordered tree. A leaf has a state coordinate and produces a symbol carrying its leaf label. An internal node produces a long list of rows, each containing its child words in their fixed order. Advancing the row index deletes a few rows at one end and inserts a few at the other, so it is inexpensive. A distinguished second shift must instead separate the words even when an alignment crosses row boundaries.

The shared alignment lemma in Section 2 handles this difficulty. Leaf labels preserve child indices, not row indices. An unsplit source row loses symbols because its child words are separated from the corresponding target words. A split source row forces a long unmatched target interval between consecutive matches. The intervals charged by different source rows are disjoint. Thus separation in every comparison of two rows implies separation of the whole words, without assuming that row boundaries match.

The two constructions arrange the shifts differently. In the finite tree of Section 3, a child’s period is its parent’s period times a new prime. At an internal node, a vector u_v advances rows and a multiple w_v has the parent period. The distinct prime factors prevent a difference of row indices from cancelling the distinguished shift in too many children. This gives a deterministic separation proof on a finite group. For a Fourier character, activity at a node with few active children has a short period and is detected by an average of inexpensive row shifts. A character escaping every such test must be active on many leaves; individual leaf changes then bound its contribution. The section closes with binary witnesses for every large cap.

The circle tree in Sections 4 and 5 uses quantized phases instead of finite residues. Random prime lists and coefficients make the root half-turn separate every base phase. The probabilistic assertion is only for one fixed shift; the all-phase conclusion follows deterministically from the alignment lemma. In its Fourier argument, tiny rotations detect large integer frequency sums and rational row shifts detect failed prime congruences. When both tests fail, divisibility forces every nonzero sum to continue through many children. This supplies the second binary proof without

using the finite tree.

Both routes use the same elementary delimiter code from Section 2. Its distortion loss grows with the block width, but that width is only $O(k \log k)$ here and does not change the logarithmic lower-bound scale. Appendix A proves a stronger, separate alphabet-conversion theorem: for words of one prescribed length M over at most M letters, blocks of width M^2 preserve edit distance up to scale and an absolute factor 804. Its proof is independent of both trees, and neither headline lower bound requires this stronger conversion.

2 Edit deficits, ordered rows and binary transfer

Both trees need four elementary tools. Subsequence deficits and an alignment lemma turn separation of child blocks into separation of whole words, even when matches cross row boundaries. A cut representation permits the later Fourier estimates, and a prime count supplies the periods. Finally, a short binary code transfers either tree's obstruction to binary words with a controlled width loss.

2.1 Subsequence matchings

A common subsequence matching of words $x = x_1 \cdots x_a$ and $y = y_1 \cdots y_b$ is a list of pairs $(i_1, j_1), \dots, (i_s, j_s)$ whose two index lists are strictly increasing and for which $x_{i_t} = y_{j_t}$ for every t . Its largest possible size is $\text{LCS}(x, y)$. The insertion–deletion distance is

$$\Delta(x, y) = |x| + |y| - 2 \text{LCS}(x, y).$$

Indeed, symbols of x that survive an insertion–deletion script form a common subsequence; conversely, retaining a longest common subsequence and deleting and inserting the remaining symbols attains this value. Replacing each substitution by one deletion and one insertion gives

$$\text{ED}(x, y) \leq \Delta(x, y) \leq 2 \text{ED}(x, y). \quad (1)$$

For words of equal length n , put $r_n(x, y) = n - \text{LCS}(x, y)$. Thus

$$r_n(x, y) \leq \text{ED}(x, y) \leq 2r_n(x, y). \quad (2)$$

A matching of size $n - E$ between two length- n words leaves exactly E unmatched positions on each side.

2.2 The common alignment lemma

Suppose each row consists of one block from each of several pairwise disjoint alphabets, in the same order. An equal-symbol match must then join blocks with the same index, although their row indices can differ. Products of words in the next lemma mean concatenation in increasing index order.

Lemma 2.1 (Rows with ordered tags). *Let $T, B \geq 1$ and $m \geq 2$ be integers, and let $\Gamma_1, \dots, \Gamma_m$ be pairwise disjoint alphabets. For $1 \leq a, b \leq T$ and $1 \leq i \leq m$, take $X_{a,i}, Y_{b,i} \in \Gamma_i^B$, and define*

$$X = \prod_{a=1}^T (X_{a,1} \cdots X_{a,m}), \quad Y = \prod_{b=1}^T (Y_{b,1} \cdots Y_{b,m}), \quad N = TmB.$$

Suppose $d \geq 0$ is real and

$$\sum_{i=1}^m r_B(X_{a,i}, Y_{b,i}) \geq d \quad (1 \leq a, b \leq T). \quad (3)$$

With $g = (m-1)B$, one has

$$r_N(X, Y) \geq \frac{Tdg}{d+g} \geq \frac{T}{2} \min(d, g). \quad (4)$$

In particular, $d \leq g$ implies $r_N(X, Y) \geq Td/2$.

Proof. Fix an increasing matching and let E be its unmatched count on either side. Call a source row split if it has matches in at least two target rows, and let s be the number of split source rows.

A source row that is not split and has matches sends them all into one target row, say b . The disjoint alphabets confine matches from its block i to block i in that target row. The restriction of the matching to those two blocks has size at most their LCS, even if other source rows also use the target block. Condition (3) therefore leaves at least d unmatched source positions in this row. A source row with no matches loses all mB positions, again at least d , since (3) implies $d \leq mB$. The nonsplit rows are disjoint, so

$$E \geq d(T-s). \quad (5)$$

For a split source row, list its matches in order and choose two consecutive entries at a transition between target rows. They are also consecutive in the entire matching: every source position between them lies in the same source row, so an intervening global match would have appeared in its list. Let i, i' be their source block indices. Their order gives $i \leq i'$, and disjoint alphabets give the same two block indices in the target. Between their target positions lie at least

$$(m-i+i'-1)B \geq (m-1)B = g$$

positions in complete blocks. This is the count for adjacent target rows; intervening rows only increase it. All positions in that open target interval are unmatched, because the selected matches are globally consecutive. The intervals selected for different source rows are disjoint by strict monotonicity of the matching. Hence

$$E \geq sg. \quad (6)$$

Since $g > 0$, (6) gives $s \leq E/g$. Substitute this in (5) to obtain $E \geq d(T - E/g)$, and rearrange: $E \geq Tdg/(d+g)$. If $d \leq g$, then $dg/(d+g) \geq d/2$; if $g \leq d$, it is at least $g/2$. These bounds hold for every matching, including when $d = 0$ or $T = 1$, and therefore for a longest one. $\square$

The two inequalities in the proof bound the same number E using different words: nonsplit rows give source losses, and split rows give target losses. Equal total lengths make the unmatched counts equal. The fixed order, equal block lengths and disjoint tags supply the target gaps; no alignment of row boundaries is assumed. Figure 1 illustrates the gap created by a split row.

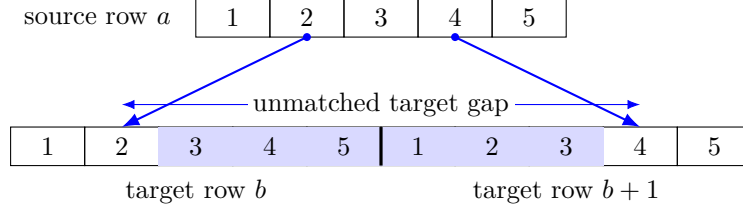


Figure 1: A split source row in Lemma 2.1, with $m = 5$. The two arrows are consecutive matches from source blocks $i = 2$ and $i' = 4$. Their target positions lie in different rows, so all positions between them are unmatched. The six shaded complete blocks contribute $(m - i + i' - 1)B = 6B \geq (m - 1)B$ positions. The diagram is schematic; target rows further apart would only enlarge the gap.

2.3 Cuts and prime counts

We record two background facts for both constructions. The first turns an ℓ_1 distance on a finite set into a nonnegative sum of binary separation tests. This standard cut description is stated, for example, in Section 4 of Linial, London and Rabinovich [12]; the proof below includes a summability bound for maps with infinitely many coordinates.

Lemma 2.2 (Finite cut representation). *Let S be finite and $f : S \rightarrow \ell_1$. There are nonnegative finite weights a_C , indexed by nonempty proper subsets $C \subset S$, such that*

$$\|f(x) - f(y)\|_1 = \sum_{\emptyset \neq C \subsetneq S} a_C |\mathbf{1}_C(x) - \mathbf{1}_C(y)|.$$

Moreover,

$$\sum_{\emptyset \neq C \subsetneq S} a_C \leq \sum_{\{x,y\} \subset S, x \neq y} \|f(x) - f(y)\|_1,$$

where the sum on the right is over unordered pairs.

Proof. For each real coordinate f_j ,

$$|f_j(x) - f_j(y)| = \int_{\mathbb{R}} |\mathbf{1}_{\{f_j > t\}}(x) - \mathbf{1}_{\{f_j > t\}}(y)| dt.$$

Discard empty and full level sets, which contribute zero, then sum over coordinates and group the remaining level sets by their subset of S . All terms are nonnegative, so monotone convergence permits this grouping. Every nontrivial cut separates at least one unordered pair. Summing the identity over those pairs bounds the total weight of the nontrivial cuts by the stated finite sum. $\square$

The second fact supplies all primes used below. We use the elementary binomial-coefficient argument, following the classical prime-product method found, for example, in Erdős [4]. Write $\pi(x)$ for the number of primes at most x .

Lemma 2.3 (Prime supply). *There is an absolute $c > 0$ such that $\pi(x) \geq cx/\log x$ for all sufficiently large real x . Consequently, for all sufficiently large integers k , the interval $[k^4, k^5]$ contains at least $20k^2$ primes and $[k^2, k^3]$ contains more than $1600k$ primes.*

Proof. For an integer $a \geq 1$, the largest coefficient in the expansion of $(1 + 1)^{2a}$ is the central one, so $\binom{2a}{a} \geq 4^a/(2a + 1)$. The exponent of a prime p in this coefficient is

$$\sum_{s \geq 1} \left(\left\lfloor \frac{2a}{p^s} \right\rfloor - 2 \left\lfloor \frac{a}{p^s} \right\rfloor \right).$$

Each summand is zero or one and vanishes for $p^s > 2a$. Hence the whole p -power dividing $\binom{2a}{a}$ is at most $2a$, and $\binom{2a}{a} \leq (2a)^{\pi(2a)}$. Taking logarithms gives $\pi(2a) \geq c(2a)/\log(2a)$ for large a . Taking $a = \lfloor x/2 \rfloor$ yields the asserted lower bound for real x . Finally subtract $\pi(k^4) \leq k^4$ from the lower bound at k^5 , and $\pi(k^2) \leq k^2$ from the lower bound at k^3 . The respective differences eventually exceed $20k^2$ and $1600k$. $\square$

2.4 A short binary code

The trees initially use leaf labels and hence a growing alphabet. The following code will suffice for both binary lower bounds. It preserves subsequence deficits from below, while its expansion is at most its block width. For the tree alphabets that width will be $O(k \log k)$, small compared with their eventual distortion lower bounds.

Lemma 2.4 (Delimiter transfer). *Let Γ be a finite alphabet of size $A \geq 2$, let $t \geq \lceil \log_2 A \rceil$ be an integer, and put $B = 2t + 3$. Give the symbols of Γ distinct labels $(a_1, \dots, a_t) \in \{0, 1\}^t$. Let $c(a)$ be the prefix 110 followed, in order, by the two-bit words $a_j 0$ for $j = 1, \dots, t$. Let $\mathcal{B}$ extend c by concatenation. For every integer $n \geq 0$ and every $x, y \in \Gamma^n$,*

$$\frac{1}{2}r_n(x, y) \leq \text{ED}(\mathcal{B}(x), \mathcal{B}(y)) \leq B \text{ED}(x, y). \quad (7)$$

Proof. The pattern 11 occurs in a concatenation only at block starts. The final zero of each block prevents an additional occurrence across a boundary. Fix a longest common subsequence matching of the encoded words, and let l be its unmatched count on each side.

At most l source blocks contain an unmatched source bit. Among the fully matched source blocks, consider one whose image is not a consecutive target interval. Two consecutive bits of that source block then have matched target positions with a nonempty gap. No target position in that gap is matched: there is no source position between the two consecutive source bits. The gaps chosen for different source blocks are disjoint by monotonicity. Thus at most l additional source blocks have a nonconsecutive target image.

Every remaining source block is matched to a consecutive target interval of length B . Its initial 11 starts at a target block boundary, so the entire block equals that target block. These target blocks are distinct and occur in increasing order. Their original letters form a common subsequence of x, y with at least $n - 2l$ letters. Hence $r_n(x, y) \leq 2l$. By (2), encoded edit distance is at least its one-sided deficit l , proving the lower bound. Simulating each original symbol edit by at most B bit edits gives the upper bound, with no restriction on intermediate lengths. When $n = 0$, both sides are zero. $\square$

By (2), the lower bound in (7) is at least $\text{ED}(x, y)/4$. Consequently, if a finite set of equal-length words has ℓ_1 distortion at least D_0 , its encoded image has distortion at least $D_0/(4B)$: composing an embedding of the encoded image with $\mathcal{B}$ incurs at most the factor $4B$.

3 A finite tree with nested periods

We first construct words over a finite alphabet that grows with the tree. At every internal node, one vector advances the rows cheaply and a multiple of it has a prescribed smaller period. The latter vector separates words for every state. We then use Fourier analysis to show that an ℓ_1 map cannot preserve both kinds of displacement with small distortion. The short binary code from Lemma 2.4 will then complete the first proof of Theorem 1.1.

3.1 Periods and word construction

Fix a sufficiently large integer k and put

$$m = 20k, \quad T = k^{20k}. \quad (8)$$

Use the full ordered m -ary tree of depth k , with root o at depth zero and children $v_1, \dots, v_m$ at each internal node v . By Lemma 2.3, choose primes $p_{j,i} \in [k^4, k^5]$ distinct over all pairs $0 \leq j < k$, $1 \leq i \leq m$. Nodes at the same depth use the same list. Define their periods by

$$Q_o = 2, \quad Q_{vi} = Q_v p_{j,i} \quad \text{when } v \text{ has depth } j.$$

All these primes are odd for large k .

On the finite abelian group

$$A = \prod_{u \text{ leaf}} \mathbb{Z}/Q_u \mathbb{Z}$$

define vectors supported on their respective subtrees:

$$\begin{aligned} w_u &= e_u & (u \text{ leaf}), \\ u_v &= \sum_{i=1}^m w_{vi}, & R_v = \prod_{i=1}^m p_{j,i}, & w_v = R_v u_v \quad (v \text{ internal at depth } j). \end{aligned} \quad (9)$$

Here e_u increments the coordinate of leaf u .

Lemma 3.1 (Orders of the node vectors). *For every node v , $\text{ord}(w_v) = Q_v$. At every internal node,*

$$\text{ord}(u_v) = Q_v R_v, \quad \gcd(Q_v, R_v) = 1.$$

Proof. At a leaf, e_u has order Q_u . Suppose the assertion holds at the children of an internal node v . Their vectors have disjoint supports and orders $Q_v p_{j,i}$, so their sum has order

$$\text{ord}(u_v) = \text{lcm}_{1 \leq i \leq m} (Q_v p_{j,i}) = Q_v R_v.$$

The primes dividing Q_v come from earlier depths, together with two, and none divides R_v . Multiplication by R_v therefore leaves a vector of order Q_v , namely w_v . $\square$

For $x \in A$, the leaf word is the single symbol $G_u(x) = (u, x_u)$. At an internal node, define

$$G_v(x) = \prod_{a=0}^{T-1} (G_{v1}(x + au_v) \cdots G_{vm}(x + au_v)), \quad (10)$$

where the product concatenates the rows in increasing a . Only leaf coordinates below v affect G_v . At depth j and height $h = k - j$, its length is $L_v = (mT)^h$. Write

$$M = (mT)^k, \quad G = G_o.$$

Every symbol retains its leaf label. Thus the alphabets occurring below different children of a node are disjoint, exactly as required by Lemma 2.1.

3.2 Cheap translations and uniform separation

We first bound the cost of advancing rows or changing one leaf. These bounds hold at every state and will later be averaged over A .

Lemma 3.2 (Inexpensive finite translations). *For every leaf u , every $x \in A$ and every integer t ,*

$$\text{ED}(G(x), G(x + te_u)) \leq M/m^k.$$

If v is internal at depth j , then for every $x \in A$ and integer $1 \leq s \leq T$,

$$\text{ED}(G(x), G(x + su_v)) \leq \frac{2s}{T} \frac{M}{m^j}. \quad (11)$$

Proof. A leaf occurs $T^k = M/m^k$ times in the root word. Altering that coordinate can change only those symbols, so substitutions prove the first bound.

Within one copy of G_v , translation by su_v changes the row parameters from $0, \dots, T-1$ to $s, \dots, T-1+s$. The two lists share $T-s$ rows. Deleting the first s rows on one side and inserting the last s on the other costs at most $2sL_v/T$ edits. A node at depth j occurs T^j times in G , with total length $T^j L_v = M/m^j$. Each occurrence has an input offset contributed by ancestors. That offset is the same on both sides and commutes with su_v , so the local script applies in every occurrence. The word is unchanged outside them. Summing the costs proves (11). $\square$

We now prove separation for every nonzero residue along the order- Q_v vector $w_v = R_v u_v$, uniformly in the initial state.

Proposition 3.3 (Separation in the finite tree). *For every node v of height h , every $x \in A$, and every integer t with $Q_v \nmid t$,*

$$r_{L_v}(G_v(x), G_v(x + tw_v)) \geq 4^{-h} L_v. \quad (12)$$

Proof. At a leaf, the two residue symbols differ. Suppose the proposition holds at the children of an internal node v , and write $B = L_v/(mT)$ for their common word length.

Compare any source row a in $G_v(x)$ and any target row a' in $G_v(x + tw_v)$. In child vi , the relative input shift is

$$(a' - a + tR_v)w_{vi}.$$

At least $m/2$ children have $Q_{vi} \nmid a' - a + tR_v$. If $a' = a$, the assertion holds for every child: $Q_v \nmid t$ and $\gcd(Q_v, R_v) = 1$ imply $Q_v \nmid tR_v$. If $a' \neq a$, divisibility by Q_{vi} would require $p_{j,i} \mid a' - a$, since $p_{j,i} \mid R_v$. But

$$0 < |a' - a| < T = k^{20k} \quad \text{and} \quad (k^4)^{m/2} = k^{40k} > T.$$

Thus $m/2$ distinct primes of the required sizes cannot all divide $a' - a$.

By induction, the sum of the deficits over the child blocks in every pair of rows is at least

$$d_h = \frac{m}{2} 4^{-(h-1)} B. \quad (13)$$

The child blocks have equal length, fixed order and disjoint alphabets. Also $d_h \leq (m-1)B$. Lemma 2.1 therefore gives

$$r_{L_v}(G_v(x), G_v(x + tw_v)) \geq \frac{Td_h}{2} = 4^{-h} mTB = 4^{-h} L_v.$$

The lemma applies to arbitrary matchings, so this proves the induction for every state x . $\square$

We have obtained the two geometric facts needed for the obstruction: the root shift w_o is far at every state, while the shifts in Lemma 3.2 are cheap. The remaining argument compares their average images under an arbitrary ℓ_1 embedding.

3.3 Fourier detection of sparse activity

For a finite abelian group A , use uniform probability measure and write $\hat{A}$ for its character group. The following identity represents an averaged ℓ_1 displacement through cuts and Fourier coefficients. It is the elementary translation principle underlying the comparison with the cut methods of Khot–Naor and Krauthgamer–Rabani [8, 10].

Lemma 3.4 (Finite-group displacement). *If $F : A \rightarrow \ell_1$ is any map on a finite abelian group, then there are weights $H(\chi) \geq 0$ such that, for every $z \in A$,*

$$\mathbb{E}_x \|F(x+z) - F(x)\|_1 = \sum_{\chi \in \hat{A}} |1 - \chi(z)|^2 H(\chi). \quad (14)$$

The map F need not be injective.

Proof. Apply Lemma 2.2 to the finite image of F and pull each cut back to an indicator $U : A \rightarrow \{0, 1\}$. Since an indicator difference has absolute value equal to its squared absolute value, Parseval gives

$$\mathbb{E}_x |U(x+z) - U(x)| = \sum_{\chi \in \hat{A}} |\hat{U}(\chi)|^2 |1 - \chi(z)|^2, \quad \hat{U}(\chi) = \mathbb{E}_x U(x) \overline{\chi(x)}.$$

Multiply by the cut weights and add. The coefficient of each multiplier is the nonnegative number $H(\chi) = \sum_C a_C |\hat{U}_C(\chi)|^2$. $\square$

For a character χ , call a node v active when $\chi(w_v) \neq 1$. Let $\text{wt}(\chi)$ be the number of active leaves. The period design has one useful consequence: activity at a node with few active children is detected by a short average of cheap shifts. Frequencies that escape all these tests must have large leaf support.

Proposition 3.5 (Finite obstruction). *For all sufficiently large k , the word image $\{G(x) : x \in A\}$, with ordinary edit distance, has ℓ_1 distortion at least $C_1^{-1}(k/4)^k$, for an absolute constant C_1 .*

Proof. Take an embedding of the word image with distortion D and rescale it so that its induced distance ρ on parameter states satisfies

$$\frac{1}{D} \text{ED}(G(x), G(y)) \leq \rho(x, y) \leq \text{ED}(G(x), G(y)).$$

The map $x \mapsto G(x)$ need not be injective; composing it with the embedding still gives a map to which Lemma 3.4 applies. Write

$$V(z) = \mathbb{E}_x \rho(x, x+z) = \sum_{\chi} |1 - \chi(z)|^2 H(\chi), \quad H(\chi) \geq 0.$$

First average over the entire cyclic coordinate of one leaf u . The average of $|1 - \chi(te_u)|^2$ is two if χ is nontrivial on that coordinate and zero otherwise. Sum the leaf-cost bound in Lemma 3.2 over all m^k leaves. It follows that

$$2 \sum_{\chi} \text{wt}(\chi) H(\chi) \leq M. \quad (15)$$

Now suppose an internal node v at depth j is active for χ but has fewer than k active children. Since $\chi(w_v) = \chi(u_v)^{R_v} \neq 1$, the value $\chi(u_v)$ is a nontrivial root of unity. The inactive children

contribute one to $\chi(u_v) = \prod_i \chi(w_{vi})$, so at least one child is active. Let $q = \text{ord}(\chi(u_v))$. This order divides the least common multiple of the active child periods, and

$$q \leq Q_v \prod_{i: v_i \text{ active}} p_{j,i} \leq 2k^{5j+5(k-1)} \leq 2k^{10k}.$$

Put $J = 4k^{10k}$. For large k , $J \leq T$ and $q \leq J/2$. The integers $1, \dots, J$ contain $\lfloor J/q \rfloor$ complete periods of this root of unity, with total length at least $J/2$. On a complete nontrivial period the mean of $|1 - \chi(su_v)|^2$ is two. Consequently

$$\frac{1}{J} \sum_{s=1}^J |1 - \chi(su_v)|^2 \geq 1.$$

This uses complete periods inside the averaging interval and does not require q to divide J .

Because all Fourier weights are nonnegative, the total weight of characters with this property at the fixed node v is at most $J^{-1} \sum_{s=1}^J V(su_v)$. Lemma 3.2 bounds that average by $(2J/T)M/m^j$. There are m^j nodes at depth j . Summing over nodes and all k internal depths bounds the weight of characters having any active internal node with fewer than k active children by

$$2k(J/T)M. \tag{16}$$

The sets of characters for different nodes may overlap; nonnegativity makes their sum an upper bound for the union.

Every other character active at the root has at least k active children at each active internal node. Iterating down the tree gives $\text{wt}(\chi) \geq k^k$. By (15), these characters have total weight at most $M/(2k^k)$. Root-inactive characters contribute zero to $V(w_o)$, and any root multiplier is at most four. Hence

$$V(w_o) \leq 4M \left(\frac{2kJ}{T} + \frac{1}{2k^k} \right) \leq C_1 M k^{-k}. \tag{17}$$

Here $J/T = 4k^{-10k}$, so an absolute C_1 suffices for all large k .

Finally $Q_o = 2$. Proposition 3.3 with $t = 1$ and (2) imply $\rho(x, x + w_o) \geq 4^{-k} M/D$ at every state. Its average, compared with (17), gives $D \geq C_1^{-1} (k/4)^k$. $\square$

3.4 Binary witnesses for every sufficiently large cap

The finite tree alphabet contains one symbol for each leaf and each residue in its coordinate group. Since $Q_u \leq 2k^{5k}$ at every leaf, its size K satisfies

$$K \leq 2m^k k^{5k} = 2(20k)^k k^{5k}.$$

Choose

$$t_k = \left\lceil \log_2(2(20k)^k k^{5k}) \right\rceil, \quad B_k = 2t_k + 3 = O(k \log k).$$

There are enough t_k -bit labels for all symbols, so Lemma 2.4 applies with width B_k . Its distortion loss is at most $4B_k$. Hence Proposition 3.5 supplies a finite binary witness of the single length

$$d_k = B_k M = B_k [20k \cdot k^{20k}]^k \tag{18}$$

whose distortion is at least

$$D_k := \frac{(k/4)^k}{4C_1 B_k}.$$

The separated root pair and the lower bound in Lemma 2.4 ensure that the witness contains at least two words.

The explicit widths B_k and lengths M increase with k , so d_k increases to infinity, independently of which admissible primes were chosen. Moreover,

$$\begin{aligned}\log d_k &= 20k^2 \log k + k \log(20k) + \log B_k \\ &= 20k^2 \log k + O(k \log k), \\ \log D_k &\geq c' k \log k\end{aligned}$$

for an absolute $c' > 0$ and all sufficiently large k . Given a sufficiently large integer d , choose k with $d_k \leq d < d_{k+1}$. Then $\log d \leq Ck^2 \log k$ and $\log \log d \leq C' \log k$ for absolute constants. Thus

$$\log D_k \geq c \sqrt{\log d \log \log d}.$$

The witness of length $d_k \leq d$ proves Theorem 1.1 for every sufficiently large cap.

4 A tree of circle phases

The second construction again concatenates ordered child words into rows, but a leaf now records a quantized circle phase. We will choose the increments between rows so that a half-turn of all leaf phases separates the resulting words. The choice is random; its output is one fixed word map with separation at every base phase.

4.1 Prime groups and row increments

Fix a sufficiently large integer k and set

$$g = 16k, \quad m = gk^2 = 16k^3, \quad R = k^{8k}, \quad T = k^{52k}. \quad (19)$$

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, with $|t|_{\mathbb{T}}$ denoting the circle distance from t to zero. Use a full ordered m -ary tree of depth k , and divide the children of each internal node into g groups of k^2 children. Each leaf phase will be quantized into R intervals, and each internal word will contain T rows.

Let $\mathcal{P}_k$ be the primes in $[k^2, k^3]$. Lemma 2.3 gives $|\mathcal{P}_k| > 100g$ for large k . At each internal node v , independently between nodes, choose an ordered list $r_{v,1}, \dots, r_{v,g}$ of distinct members of $\mathcal{P}_k$, uniformly without replacement. Put

$$P_v = \prod_{j=1}^g r_{v,j} \leq k^{48k}.$$

To specify the increment of a child vi in group j , choose z_{vi} uniformly from $\{0, \dots, r_{v,j} - 1\}$ and define

$$c_{vi} = 1 + z_{vi} \frac{P_v}{r_{v,j}}. \quad (20)$$

Conditionally on all prime lists, all these residues are independent. The increment c_{vi}/P_v has the decomposition

$$\frac{c_{vi}}{P_v} = \frac{1}{P_v} + \frac{z_{vi}}{r_{v,j}}.$$

Thus every child has a common drift of $1/P_v$, together with an independently chosen step on its group's prime grid. The same formula isolates the groups arithmetically: if $\ell \neq j$, then $c_{vi} \equiv 1 \pmod{r_{v,\ell}}$. At each selected prime, only coefficients from its own group can differ from one.

Associate a phase $w_u \in \mathbb{T}$ to each leaf u , and write $w \in \mathbb{T}^{m^k}$ for the full vector. For a subtree vector ω , the notation $\omega + t$ means addition of t to every coordinate of that vector. At a leaf, put

$$X_u(w_u) = (u, \lfloor R w_u \rfloor),$$

using the representative of w_u in $[0, 1)$. At an internal node, define

$$X_v(\omega) = \prod_{a=0}^{T-1} \left(X_{v1} \left(\omega_{v1} + \frac{ac_{v1}}{P_v} \right) \cdots X_{vm} \left(\omega_{vm} + \frac{ac_{vm}}{P_v} \right) \right), \quad (21)$$

where ω_{vi} is the subvector on the leaves below vi . The product concatenates rows in increasing a . At height h , the word length is $(mT)^h$. The root map $X = X_o$ has length

$$N = (mT)^k \quad (22)$$

over the alphabet $\Gamma = \{(u, b) : u \text{ a leaf, } b \in \{0, \dots, R-1\}\}$, of size $m^k R$. For fixed construction data, every position in this word quantizes one leaf phase plus a fixed offset. Hence the range of X is finite and X is measurable.

4.2 Good shifts give separation at every phase

Consider $X_v(\omega)$ and $X_v(\omega + t)$, where the scalar t is added to every leaf phase below v . In a comparison of source row a with target row a' , child vi receives the relative shift

$$t + (a' - a) \frac{c_{vi}}{P_v}.$$

The tagged-row lemma will separate the two words if enough children are separated for every row pair. Since each c_{vi} is an integer, the displayed shift depends only on $a' - a$ modulo P_v . This leads to a recursive condition on t alone, independent of the base phase. Set

$$\epsilon = R^{-1}, \quad \delta = 1/64, \quad \gamma = 1/128 = \delta/2.$$

Call a shift t eligible if $|t|_{\mathbb{T}} \geq \epsilon$.

Definition 4.1 (Good shift). At a leaf, $t \in \mathbb{T}$ is good if $|t|_{\mathbb{T}} \geq \epsilon$. At an internal node v , it is good if, for every integer a modulo P_v , at least δm children vi have $t + ac_{vi}/P_v$ good at vi .

The condition asks for at least δm children, interpreted as a real threshold on their integer count. Its deterministic consequence is the following separation estimate.

Lemma 4.2 (Conditional separation). *Fix the construction data. If a shift t is good at a node v of height h , then for every phase vector ω on its subtree,*

$$r_{(mT)^h}(X_v(\omega), X_v(\omega + t)) \geq \gamma^h (mT)^h.$$

Proof. At a leaf, two points in the same half-open interval of the partition into R intervals have circle distance strictly less than $1/R$. Thus a good shift changes the symbol for every phase, including when its distance is exactly $1/R$.

Suppose $h \geq 1$ and the statement holds at the children. Their common word length is $B = (mT)^{h-1}$. Compare any source row a and target row a' . In child vi , the relative shift is $t + (a' - a)c_{vi}/P_v$. It depends only on $a' - a$ modulo P_v , because c_{vi} is an integer. Goodness

supplies at least δm children where the induction hypothesis applies, so the sum of the deficits of the child comparisons is at least $d = \delta m \gamma^{h-1} B$.

Leaf labels give pairwise disjoint child alphabets, and the child blocks have equal length and fixed order. Since $d \leq mB/64 \leq (m-1)B$, Lemma 2.1 gives

$$r_{(mT)^h}(X_v(\omega), X_v(\omega + t)) \geq \frac{Td}{2} = \frac{\delta}{2} \gamma^{h-1} (mT)^h = \gamma^h (mT)^h.$$

This proves the induction pointwise in ω . $\square$

It remains to make the one root shift $t = 1/2$ good. We prove a statement for each fixed eligible shift that can be propagated from a child to its parent. The all-phase conclusion will then follow from the deterministic lemma.

4.3 The probability of a good fixed shift

Proposition 4.3 (Goodness of a fixed shift). *For all sufficiently large k , every node v and every fixed $t \in \mathbb{T}$ with $|t|_{\mathbb{T}} \geq \epsilon$ satisfy*

$$\mathbb{P}[t \text{ is good at } v] \geq 9/10,$$

where probability is over the construction in the subtree below v . Consequently, there exists a realization of the entire tree for which

$$r_N(X(w), X(w + 1/2)) \geq \gamma^k N \quad \text{for every } w \in \mathbb{T}^{m^k}. \quad (23)$$

Proof. The assertion is immediate at a leaf. Assume it holds at the children of an internal node v . Fix an eligible t before choosing any data at v , and suppress v from the prime notation. For a row residue a modulo P_v , let

$$S_a = \{j : a \not\equiv 0 \pmod{r_j}\}.$$

In group j , the relative shift has the form

$$t + \frac{ac_{vi}}{P_v} = t + \frac{a}{P_v} + \frac{az_{vi}}{r_j} \pmod{1}.$$

When $j \notin S_a$, the last term vanishes. When $j \in S_a$, it ranges uniformly over the r_j -point grid. We therefore treat small sets S_a by choosing the prime list, and large sets S_a by using the independent child residues.

For the first case, we show that the parent list has probability at least $1 - g/|\mathcal{P}_k|$ of satisfying

$$|t + a/P_v|_{\mathbb{T}} \geq \epsilon \quad \text{for every } a \text{ with } |S_a| \leq k. \quad (24)$$

Because P_v is squarefree, the reduced denominator of such a fraction a/P_v is a product of at most k selected primes, and is therefore at most $K_0 = k^{3k}$. Distinct rationals on the circle with denominators at most K_0 have circle distance at least $1/K_0^2$. Since $2\epsilon = 2k^{-8k} < 1/K_0^2$, at most one of all such rationals lies at distance strictly less than ϵ from $-t$.

This candidate rational, if it exists, is determined by t before the list is sampled. It is nonzero because $|t|_{\mathbb{T}} \geq \epsilon$, so its reduced denominator has a prime factor. Fix one such factor. Unless this prime belongs to the sampled list, the candidate cannot equal any of the fractions under consideration. A prime outside $\mathcal{P}_k$ is never selected; a prime inside the pool is selected with probability exactly $g/|\mathcal{P}_k|$. This proves the claimed probability of (24).

Now fix a parent prime list satisfying (24). The remaining random data for different children are independent: each consists of its residue z_{vi} and its entire child subtree. Fix one row residue a modulo P_v .

If $|S_a| \leq k$, a child in a group $j \notin S_a$ sees the shift

$$t + \frac{ac_{vi}}{P_v} = t + \frac{a}{P_v} \pmod{1},$$

since $r_j \mid a$. It is eligible by (24). By induction, the children in these groups independently have the required goodness with probability at least $9/10$. There are at least $(g - k)k^2$ such children.

If $|S_a| > k$, use children in groups $j \in S_a$, of which there are at least $k^3 = m/16$. Multiplication by a permutes the residues modulo r_j , so each relative shift is uniform on a translate of the r_j -point grid. An arc of length 2ϵ contains a fraction at most $2\epsilon + 1/r_j$ of that grid. The shift is therefore eligible with probability at least $1 - 2\epsilon - 1/r_j$. Conditional on any eligible value, the independent child subtree is good with probability at least $9/10$. Each indicated child succeeds with probability at least $4/5$ for large k , and these success events are independent.

In either case, select $n = m/16$ independent indicated children. Each has success probability at least $4/5$, and $\delta m = m/64 = n/4$. Hoeffding's inequality for independent bounded variables [7, Theorem 2] gives

$$\mathbb{P}[\text{fewer than } \delta m \text{ successes}] \leq \exp\left(-2n \left(\frac{4}{5} - \frac{1}{4}\right)^2\right) = \exp(-121m/3200).$$

The variables need not have identical distributions. A union bound over the $P_v \leq k^{48k}$ possible values of a , followed by the probability of failure of the parent event, gives

$$\mathbb{P}[t \text{ is not good at } v] \leq \frac{g}{|\mathcal{P}_k|} + \exp(48k \log k - 121m/3200) < \frac{1}{10}$$

for all sufficiently large k . Independence between different residues a is unnecessary. This proves the induction.

Apply the statement to the one fixed root shift $t = 1/2$, which is eligible for large k . Choose a realization in which it is good. Lemma 4.2 then gives (23) for every base phase. No simultaneous random choice for all eligible shifts is asserted. $\square$

Fix this realization for the rest of the torus argument. The prime lists and coefficients now have fixed values. Only their stated size and congruence properties will be used below.

5 Torus displacements, Fourier detection and binary transfer

The fixed construction separates every phase vector from its half-turn translate. We now compare this displacement with whole-row shifts, tiny rotations, and changes of a single leaf. The Fourier argument will use the cutoff

$$Q = k^{12k}.$$

The ratio $P_v/T \leq k^{-4k}$ controls the cost of whole-row shifts, and $R/Q = k^{-4k}$ controls the cost of tiny rotations. After proving the comparison for an arbitrary finite-range map on the torus, we apply the shared binary code.

5.1 Pointwise and averaged edit costs

For a node v , let $e_v \in \mathbb{R}^{m^k}$ have entry one on the leaves below v and zero elsewhere. At an internal node, define

$$h_v(a) = \frac{a}{P_v} \sum_i c_{vi} e_{vi} \pmod{1} \quad (a \in \mathbb{Z}/P_v\mathbb{Z}). \quad (25)$$

This is well defined because each c_{vi} is an integer. Let q be the depth of v .

Lemma 5.1 (Inexpensive torus translations). *For the realization fixed after Proposition 4.3, the following estimates hold, with q denoting the depth of v in the first two statements.*

(i) *If v is internal and a is an integer with $0 \leq a < P_v$, then for every w ,*

$$\text{ED}(X(w), X(w + h_v(a))) \leq 2Nm^{-q} \frac{P_v}{T}. \quad (26)$$

(ii) *If v is any node and $0 \leq \eta \leq 1/Q$, then*

$$\mathbb{E}_w \text{ED}(X(w), X(w + \eta e_v)) \leq Nm^{-q} \frac{R}{Q}, \quad (27)$$

where w has normalized Haar measure on $\mathbb{T}^{m^k}$.

(iii) *If u is a leaf, then for every w and $\eta \in \mathbb{T}$,*

$$\text{ED}(X(w), X(w + \eta e_u)) \leq Nm^{-k}. \quad (28)$$

Proof. A node at depth q occurs T^q times in the root word, each time in a word of length $(mT)^{k-q}$. Its occurrences occupy the fraction m^{-q} of all positions. Every occurrence receives its subtree phase vector plus fixed offsets from its ancestors, and those offsets are the same on both sides of each comparison.

Within an occurrence of X_v , addition of $h_v(a)$ changes the row indices from $0, \dots, T-1$ to $a, \dots, T-1+a$. Using the representative $0 \leq a < P_v \leq k^{48k} < T$, the lists share $T-a$ rows. Deleting a rows and inserting a rows costs at most $2a/T$ times the occurrence length. Nothing changes outside the occurrences of v . This proves (26); it requires no divisibility of T by P_v .

For a tiny rotation, each affected symbol quantizes a uniformly distributed leaf phase plus a fixed offset. A rotation by $0 \leq \eta \leq 1/Q$ changes its interval with probability $R\eta \leq R/Q$. Sum these positional mismatch probabilities and use substitutions to prove (27). Linearity of expectation suffices even when different occurrences use the same phase. This estimate is an average over w . Finally, an arbitrary change of one leaf can affect only its Nm^{-k} occurrences, proving (28) pointwise. $\square$

5.2 The Fourier identity on the torus

The next identity expresses averaged ℓ_1 distances through nonnegative frequency weights. It is the continuous counterpart of Lemma 3.4; its averaging measure and its integer frequencies are different.

Lemma 5.2 (Torus displacement). *Let $n \geq 1$ be an integer and let $F : \mathbb{T}^n \rightarrow \ell_1$ be a measurable map with finite range. Set*

$$L_F(h) = \mathbb{E}_w \|F(w+h) - F(w)\|_1 \quad (h \in \mathbb{T}^n),$$

using normalized Haar measure. There are summable coefficients $\mu_\xi \geq 0$, indexed by $\xi \in \mathbb{Z}^n$, such that

$$L_F(h) = \sum_{\xi \in \mathbb{Z}^n} \mu_\xi (1 - \cos(2\pi \langle \xi, h \rangle)) \quad (h \in \mathbb{T}^n). \quad (29)$$

Proof. Apply Lemma 2.2 to the finite range of F and pull a cut back to a measurable indicator U on $\mathbb{T}^n$. It belongs to $L^2(\mathbb{T}^n)$. For $\widehat{U}(\xi) = \int_{\mathbb{T}^n} U(w) e^{-2\pi i \langle \xi, w \rangle} dw$, Parseval and the Fourier translation rule give

$$\int_{\mathbb{T}^n} |U(w+h) - U(w)|^2 dw = \sum_{\xi \in \mathbb{Z}^n} 2|\widehat{U}(\xi)|^2 (1 - \cos(2\pi \langle \xi, h \rangle)).$$

An indicator difference has absolute value equal to its squared absolute value. Multiply the identity by each cut weight and sum over the finitely many cut types. The resulting coefficients are nonnegative. Their sum is at most twice the total cut weight, because $\sum_\xi |\widehat{U}(\xi)|^2 = \int |U|^2 \leq 1$. This also proves summability and justifies the identity for every h . $\square$

5.3 Two tests and forced branching

Apply the notation of Lemma 5.2 with $n = m^k$. For a frequency $\xi \in \mathbb{Z}^{m^k}$ and a node v , define its integer summary

$$s_v = \langle \xi, e_v \rangle = \sum_{u \text{ leaf below } v} \xi_u.$$

Thus $s_v = \sum_i s_{vi}$ at an internal node and $s_u = \xi_u$ at a leaf. The multiplier of the root half-turn is $1 - \cos(\pi s_o)$, equal to two for odd s_o and zero for even s_o . We bound it using tests that cover every frequency with $s_o \neq 0$.

Lemma 5.3 (Torus detection inequality). *For the fixed construction and its shifts $h_v(a)$, every measurable finite-range $F : \mathbb{T}^{m^k} \rightarrow \ell_1$ satisfies*

$$\begin{aligned} L_F(e_o/2) &\leq 4 \sum_{v \text{ all nodes}} \mathbb{E}_{\eta \in [0, 1/Q]} L_F(\eta e_v) + 2 \sum_{v \text{ internal}} \mathbb{E}_{a \in \mathbb{Z}/P_v \mathbb{Z}} L_F(h_v(a)) \\ &\quad + 2k^{-k} \sum_{u \text{ leaf}} \mathbb{E}_{\eta \in \mathbb{T}} L_F(\eta e_u). \end{aligned} \quad (30)$$

Every interval, finite-group and circle average has total mass one.

Proof. We compare the multipliers of a fixed frequency in (29). If $|s_v| \geq Q$ at some node, then

$$Q \int_0^{1/Q} (1 - \cos(2\pi s_v \eta)) d\eta = 1 - \frac{Q}{2\pi s_v} \sin(2\pi s_v/Q) \geq 1 - \frac{1}{2\pi} \geq \frac{1}{2}.$$

The first sum in (30), with coefficient four, therefore dominates the root multiplier, which is at most two.

If an internal node satisfies

$$\sum_i c_{vi} s_{vi} \not\equiv 0 \pmod{P_v}, \quad (31)$$

then the average of the multiplier for $h_v(a)$ over $a \in \mathbb{Z}/P_v \mathbb{Z}$ is exactly one, by the sum of a nontrivial cyclic character. The second sum, with coefficient two, dominates the root multiplier in this case.

It remains to consider a frequency with $|s_v| < Q$ at every node and with every internal congruence equal to zero. If $s_o = 0$, its root multiplier vanishes. Suppose $s_o \neq 0$, and consider an internal node v with $s_v \neq 0$. Call a group of children empty if all summaries in that group vanish. For an

empty group j , reduce the zero congruence modulo $r_{v,j}$. The coefficients outside that group are one modulo $r_{v,j}$ and the group contributes zero, so $r_{v,j} \mid s_v$.

If at least $g/2 = 8k$ groups were empty, their distinct primes would have product at least $(k^2)^{8k} = k^{16k} > Q$. That product cannot divide a nonzero integer s_v with $|s_v| < Q$. Hence more than $g/2$ groups contain a child with nonzero summary, and in particular there are at least k such children. Starting from the root and repeating the conclusion gives at least k^k leaves u with $s_u \neq 0$. At each such leaf, the multiplier for ηe_u has mean one over uniform $\eta \in \mathbb{T}$. The third sum in (30), with coefficient $2k^{-k}$, is therefore at least two.

These cases bound the root multiplier for every frequency. Multiply by the nonnegative coefficients μ_ξ and sum. Their summability justifies interchange with all normalized shift averages, proving (30). $\square$

The cutoff and congruence tests have different jobs. The divisibility argument forces branching only after all summaries are known to have absolute value less than Q . It would not rule out a large nonzero integer divisible by many of the selected primes.

5.4 The binary obstruction and every length cap

Apply Lemma 2.4 to the torus alphabet of size $A = m^k R$, choosing labels of length $\lceil \log_2 A \rceil$. Write $\mathcal{B}$ for the resulting encoding. Its block width is

$$B = 2\lceil \log_2(m^k R) \rceil + 3 = O(k \log k). \quad (32)$$

The encoded image consists of binary words of one length BN . It has at least two points, since the lower code bound preserves the positive root separation in (23). We apply the detection inequality directly to an embedding of this encoded image; the code's width will enter only the upper cost estimates.

Take any embedding f of the encoded image with distortion D and rescale it to be noncontracting with expansion at most D . Set

$$F(w) = f(\mathcal{B}(X(w))), \quad L(h) = L_F(h).$$

The map F is measurable with finite range, so Lemma 5.3 applies. The pointwise separation, the three costs in Lemma 5.1, and the code bounds give

$$L(e_o/2) \geq \gamma^k N/2, \quad (33)$$

$$L(h_v(a)) \leq 2DBNm^{-q}P_v/T \quad (v \text{ internal, } a \in \mathbb{Z}/P_v\mathbb{Z}), \quad (34)$$

$$L(\eta e_v) \leq DBNm^{-q}R/Q \quad (v \text{ any node, } 0 \leq \eta \leq 1/Q), \quad (35)$$

$$L(\eta e_u) \leq DBNm^{-k} \quad (u \text{ leaf, } \eta \in \mathbb{T}). \quad (36)$$

For the second line, use the representative $0 \leq a < P_v$ in the pointwise cost proof; the displacement itself depends only on the residue. The third line uses the Haar-average cost, with no pointwise claim about tiny rotations.

There are m^q nodes at depth q . Thus the factors m^{-q} cancel when the upper bounds are summed over a depth in (30). We obtain

$$\begin{aligned} L(e_o/2) &\leq DBN \left(4(k+1) \frac{R}{Q} + 4k \frac{k^{48k}}{T} + 2k^{-k} \right) \\ &\leq C' DBN k^{-k}, \end{aligned}$$

for an absolute C' , because $R/Q = k^{-4k}$ and $k^{48k}/T = k^{-4k}$. Comparing this with (33) gives

$$D \geq \frac{(\gamma k)^k}{2C'B}, \quad \gamma = 1/128. \quad (37)$$

The denominator retains the width B from the delimiter code. Since $B = O(k \log k)$, the logarithm of the right side is at least $c_{\text{tor}} k \log k$ for an absolute $c_{\text{tor}} > 0$ and all sufficiently large k .

The binary word length satisfies

$$\log(BN) = 52k^2 \log k + O(k \log k). \quad (38)$$

For every sufficiently large integer d , put $\tau = \log d$ and choose $k = \lfloor a\sqrt{\tau/\log \tau} \rfloor$, where $a > 0$ is a sufficiently small absolute constant. There is an absolute C_2 such that $\log(BN) \leq C_2 k^2 \log k$ for all large k . Since $k \leq a\sqrt{\tau/\log \tau}$ and $\log k \leq \log \tau$ eventually, choosing $C_2 a^2 \leq 1/2$ gives $\log(BN) \leq \tau/2 < \tau$. Also, for all large τ ,

$$k \geq \frac{a}{2} \sqrt{\frac{\tau}{\log \tau}}, \quad \log k \geq \frac{1}{3} \log \tau.$$

It follows that $k \log k \geq (a/6)\sqrt{\tau \log \tau}$. The finite set $\{\mathcal{B}(X(w)) : w \in \mathbb{T}^{m^k}\} \subseteq \{0,1\}^{BN}$ therefore has distortion at least $\exp(c\sqrt{\log d \log \log d})$ and has $BN \leq d$. This completes the torus proof of Theorem 1.1 for every sufficiently large cap.

6 The uniform two-sided bound

The two lower constructions use no result from a companion manuscript. For the matching upper estimate we use the following full statement of the companion uniform upper embedding theorem [14, Theorem 8.1].

Theorem 6.1 (Uniform upper embedding, companion theorem). *There are absolute constants $C > 0$ and $d_0 \geq 3$ such that, for every finite alphabet Σ with $|\Sigma| \geq 2$ and every integer $d \geq d_0$, there are an integer $M \geq 1$ and one deterministic map $f : \Sigma^{\leq d} \rightarrow \ell_1^M$ satisfying*

$$\text{ED}(x, y) \leq \|f(x) - f(y)\|_1 \leq \exp\left(C\sqrt{\log d \log \log d}\right) \text{ED}(x, y) \quad (x, y \in \Sigma^{\leq d}).$$

Here ℓ_1^M is $\mathbb{R}^M$ with its usual ℓ_1 norm. The domain includes the empty word, and ED is ordinary unit-cost insertion, deletion and substitution distance. The dimension may depend on Σ, d ; the constants and threshold do not.

The lower inequality makes the map injective. We use only the resulting distortion bound.

Corollary 6.2. *There are absolute $c, C > 0$ and an integer d_2 such that for every integer $d \geq d_2$ and every finite alphabet Σ with $|\Sigma| \geq 2$,*

$$\exp\left(c\sqrt{\log d \log \log d}\right) \leq E_\Sigma(d) \leq \exp\left(C\sqrt{\log d \log \log d}\right). \quad (39)$$

The same bounds hold for $\sup_{\Sigma: 2 \leq |\Sigma| < \infty} E_\Sigma(d)$.

Proof. Choose two distinct symbols of Σ and identify them with zero and one. Binary edit distance is the restriction of the ambient edit distance. One inequality holds because every binary script is an ambient script. For the other, project every extra symbol of Σ to zero while fixing zero and

one. Projecting an ambient script between binary words produces a binary script of no greater cost: each operation becomes the same type of operation or no change. Thus the two distances are equal on binary words.

The witness from either proof of Theorem 1.1 is therefore an isometric subset of $\Sigma^{\leq d}$. Restricting an embedding to that subset proves the lower bound in (39). Theorem 6.1 gives the upper bound. Since the constants and thresholds are absolute, taking the supremum over finite alphabets preserves both bounds. $\square$

The alphabet may grow with d . The estimate determines the order of $\log E_\Sigma(d)$ uniformly; it leaves the constants within the exponent unspecified.

A A constant-distortion code for one prescribed length

The short code in Lemma 2.4 is sufficient for both tree lower bounds. Here we prove a stronger transfer result: a binary block code with absolute distortion for every pair of words of one prescribed length M , over an alphabet of size at most M . Its width M^2 increases the word length polynomially. The proof combines separation between segments of unequal codewords with an extraction of original common subsequences from matches between equal codewords.

Theorem A.1 (Fixed-length binary conversion). *There is an absolute integer M_0 such that, for every integer $M \geq M_0$ and every alphabet Γ with $1 \leq K = |\Gamma| \leq M$, there exists a map $c : \Gamma \rightarrow \{0, 1\}^P$ with $P = M^2$ whose concatenation extension enc satisfies*

$$\frac{P}{804} \text{ED}(x, y) \leq \text{ED}(\text{enc}(x), \text{enc}(y)) \leq P \text{ED}(x, y) \quad (x, y \in \Gamma^M). \quad (40)$$

The encoded words have length $MP = M^3$. More precisely,

$$r_{MP}(\text{enc}(x), \text{enc}(y)) \geq \frac{P}{402} r_M(x, y) \quad (x, y \in \Gamma^M). \quad (41)$$

We first choose blocks whose long segments are separated whenever the underlying letters differ. Separation of long intervals of distinct binary codewords also appears in the inner code of Schulman and Zuckerman [16, Section II-C, Condition 2 and Lemma 2]. We prove the precise property needed here directly. The later argument will handle matches between equal letters without a condition on self-overlaps of a codeword.

Lemma A.2 (Separation of unequal-label segments). *For sufficiently large integers M and integers $1 \leq K \leq M$, there are K binary words of length $P = M^2$ such that any two length- a segments of distinct codewords satisfy $\text{LCS} \leq 0.99a$ whenever*

$$a \geq a_0 := \lceil 40 \log_2(MP) \rceil = \lceil 120 \log_2 M \rceil.$$

Proof. Choose all KP bits independently and uniformly. For two fixed length- a segments in distinct codewords, put $s = \lceil 0.99a \rceil$. Every prescribed increasing matching of size s has probability 2^{-s} : its pairs use distinct independent bits on each side. A union bound over the two position lists gives

$$\mathbb{P}[\text{LCS} \geq s] \leq \binom{a}{s}^2 2^{-s} \leq 2^{-a/2}.$$

For the last estimate, let $u = a - s \leq 0.01a$. If $u = 0$, it is immediate. Otherwise $\binom{a}{u} \leq (ea/u)^u$, and $u \log_2(ea/u)$ is increasing for $0 < u < a$. Therefore

$$2 \log_2 \binom{a}{u} - s \leq 0.02a \log_2(100e) - 0.99a < -0.81a < -a/2.$$

There are at most $K^2 P^3 \leq M^2 P^3$ choices of two codewords, a segment length, and two starting positions. The probability that any comparison of length at least a_0 fails is at most

$$M^2 P^3 2^{-a_0/2} \leq M^2 P^3 (MP)^{-20} < 1.$$

For large M , $a_0 \leq P$. A realization avoiding every forbidden comparison exists. Excluding $\text{LCS} \geq \lceil 0.99a \rceil$ implies the stated weak inequality, including when $0.99a$ is an integer. $\square$

Proof of Theorem A.1. Use codewords supplied by Lemma A.2. Fix any increasing matching of N_{mat} bits between $\text{enc}(x)$ and $\text{enc}(y)$, and put $\delta = MP - N_{\text{mat}}$. Each encoded word has exactly δ unmatched positions.

We first count matches between unequal original letters. Each matched bit pair belongs to a pair (i, j) of original block positions. Along the matching, both block indices are nondecreasing. Every change of pair increases $i + j$, so at most $2M - 1$ block pairs occur. The matches of one fixed pair are consecutive in this list: any pair between two occurrences of (i, j) must have both indices equal to i, j .

Suppose a block pair with unequal letters contains a matches. Take the minimal contiguous spans containing those matched positions in the two blocks. Let their lengths be b, c , and put $g = (b - a) + (c - a)$. The spans for different block pairs are disjoint on each side, even when the pairs share one block index. Moreover, no bit matched under another pair can lie inside a span, by the consecutiveness just proved. Thus g counts globally unmatched positions, and the sum of g over all unequal-label pairs is at most 2δ .

If $a \geq a_0$, trim each span to a contiguous segment of length a . This removes g positions in total and destroys at most g matched pairs. The two segments retain at least $a - g$ matches, so Lemma A.2 gives $a - g \leq 0.99a$. Hence $g \geq 0.01a$. Summing over these large pairs bounds their total matched count by $100 \sum g \leq 200\delta$. The pairs with $a < a_0$ contribute less than $2Ma_0$ matches. Consequently the number of unequal-label matches is at most

$$200\delta + 2Ma_0. \tag{42}$$

For the equal-label matches, form a bipartite multigraph whose two vertex classes are the M source and M target block positions. Each matched bit pair with equal original letters gives one edge. Every vertex has degree at most P . The edges can be partitioned into P matchings. This is the classical bipartite edge-coloring theorem of König [9]; we derive it here from Hall's theorem. Add edges between deficient vertices, allowing parallel edges, until the graph is P -regular. The total degree deficiency is the same on the two sides, so this is possible. For any set S of left vertices, let $N(S)$ be its neighbor set. Counting incident edges gives $P|S| \leq P|N(S)|$. Hall's theorem [6, Theorem 1] supplies a perfect matching. Remove it and repeat on the remaining regular graph, then discard all added edges. This gives the required P matchings of original edges.

Within one such matching, all block indices are distinct on both sides. Since its edges came from the increasing bit matching, their block pairs are strictly increasing in both indices. Their original letters agree, so they form a common subsequence of x, y . Each of the P matchings therefore has at most $\text{LCS}(x, y)$ edges, and the total number of equal-label matches is at most $P \text{LCS}(x, y)$.

Combine the two counts with $N_{\text{mat}} = MP - \delta$:

$$Pr_M(x, y) \leq 201\delta + 2Ma_0. \quad (43)$$

Choose M_0 large enough that $2Ma_0 \leq P/2$ for every $M \geq M_0$. If $x \neq y$, then $r_M(x, y) \geq 1$, and

$$201\delta \geq Pr_M(x, y) - P/2 \geq (P/2)r_M(x, y).$$

Apply this to a longest encoded matching to obtain (41). For $x = y$, that inequality is immediate. By (2),

$$\text{ED}(\text{enc}(x), \text{enc}(y)) \geq r_{MP}(\text{enc}(x), \text{enc}(y)) \geq \frac{P}{402}r_M(x, y) \geq \frac{P}{804}\text{ED}(x, y).$$

For the upper bound, extend c to all finite words by concatenation. Simulate each insertion or deletion in an optimal original edit script by P bit operations and each substitution by at most P bit substitutions. This proves (40); the intermediate word lengths may be arbitrary. $\square$

The segment property is a property of the chosen codebook, so the theorem holds simultaneously for every pair in Γ^M . The choice is tied to M : its guarantee is for one prescribed input length and does not assert a logarithmic block width.

For the finite tree, $K \leq 2m^k k^{5k} \leq M$ for all large k . This theorem therefore also gives binary witnesses of length M^3 with distortion at least $(k/4)^k/(804C_1)$. The scale P cancels when an embedding is composed with the code. This alternative retains an absolute distortion loss, at the cost of longer words than the delimiter conversion used in Section 3.

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